Engineering Mathematics - I · Unit_04 · Unit 4

DRAFT — UE26MA141A · Unit_04 · Unit 4 · Problems on Beta and Gamma Function continued

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Problems on Beta and Gamma Function continued

1 Problems on Beta and Gamma Function continued

Study focus: Problems on Beta and Gamma Function_continued. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000058 · EV0000059

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Consider the fifth problem evaluate integral zero to two pi sine power eight theta d theta.

Key details

- This I got by using one of the properties of the definite integral, which is integral 0 to 2a f of x dx is equal to 2 times integral 0 to a f of 2a minus x dx.
- If you use this property twice, you will get this.
- And after this, to simplify, I use one of the properties of beta function we have studied earlier, which is integral 0 to pi by 2, sine power p theta, cos power q theta.
- d theta is equal to one by two times beta of p plus one divided by two, comma q plus one divided by two.
- In this case, you have P is equal to 8 and Q is equal to 0.
- That is 8 plus 1 by 2, which is 9 by 2.
- Comma q plus one by two that is zero plus one divided by two that is one by two.
- And now use the property of gamma function that is gamma of n plus 1 is equal to n into gamma n.
- And gamma 7 by 2 can be written as gamma of 5 by 2 plus 1.
- Again gamma 5 by 2 can be written as 3 by 2 plus 1 and gamma 3 by 2 can be written as gamma of 1 by 2 plus 1.
- And we know gamma half.
- Gamma of 1 by 2 is equal to square root of pi.
- So thus we can write this as gamma half whole square in the numerator because there is one more gamma half and denominator gamma 5.
- Gamma 5 is equal to 4 factory.
- Next one is evaluate integral 0 to pi by 2 square root of cot theta d theta.

- cos theta divided by sine theta whole to the power one by two d theta, which is equal to integral zero to five by two sine power minus half theta cos power half theta d theta.
- Again it is in the form now integral 0 to pi by 2 sine power p theta cos power q theta d theta, which is equal to 1 by 2 times beta of p plus 1 divided by 2.
- comma q plus 1 divided by 2.
- In this particular problem we have p is equal to minus 1 by 2 and q is equal to 1 by 2.
- Because minus half plus 1 divided by 2 is 1 by 4, 1 by 2 plus 1 divided by 2 is 3 by 4.
- But how do we simplify gamma of 1 by 4 into gamma of t by 4?

Example or application from the lecture

- Consider integral zero to two pi sine power eight theta d theta, which is equal to four times integral zero to pi by two sine power eight theta d theta.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000058 · EV0000059 · EV0000060 · +1 source units in pack.json

Core explanation · 05:00–09:59

You have studied one result which says gamma of n into gamma of one minus n is equal to pi divided by sin n pi.

Key details

- You will get gamma of one by four into gamma of one minus one by four is equal to pi divided by sine of n into pi here n is equal to one by four, so it is sine of pi by four.
- When x square is tan theta, two x dx is equal to second square theta d theta.
- Therefore, integral zero to infinity dx divided by one plus x power four is equal to integral zero to pi by two one divided by one plus tan square because x square is tan.
- x power four is tan square theta into dx is second square theta d theta divided by two into square root of tan theta.
- Observe here the limits of integration change to 0 to pi by 2.
- How we got this is you have x square is equal to tan theta.
- So when x is equal to 0, that means tan theta is equal to 0.
- Tan theta is equal to zero implies theta is equal to zero.
- And when x is equal to infinity, that means tan theta is equal to infinity and we know tan theta is infinity when theta is equal to 5 bit.
- We know 1 plus tan square theta.
- 1 plus 10 square theta is equal to second square theta.
- Second to second square theta divided by two times ten theta.
- We can observe here we can cancel second square theta.
- Integral 0 to pi by 2 d theta divided by 2 into square root of tan theta.
- That is 1 by 2 into integral 0 to pi by 2.
- 1 by root tan theta we can write it as tan theta to the power minus half.
- into d theta.
- That is equal to 1 by 2 into integral 0 to pi by 2 sin theta divided by cos theta all to the power minus h into d theta.

- Which is equal to 1 by 2 into integral 0 to pi by 2, we get it as sin power minus 1 by 2 theta into cos power plus 1 by 2 theta.
- Cos power plus 1 by 2 theta d theta.
- So again it is equal to half into half into beta of three by four, comma, one by four because we know integral 0 to pi by 2 sin power p theta cos power q theta d theta.
- is equal to 1 by 2 times beta of p plus 1 divided by 2, comma q plus 1 divided by 2.
- We had p is equal to minus 1 by 2, q is equal to 1 by 2.
- It can be also written as beta of one by four comma three by four because we know that beta of m comma n is same as.

Example or application from the lecture

- Consider the seventh problem evaluate integral zero to infinity dx divided by one plus x to the power four.
- Term x square let us take it as tan theta.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000062 · EV0000063 · EV0000064 · +1 source units in pack.json

Core explanation · 10:00–14:59

beta of n comma n so we have already proved this property while studying the properties of beta function.

Key details

- Relationship between beta gamma function so it is one by four into gamma of three by four into gamma of one by four divided by gamma of one.
- Pi divided by 1 by square root of 2 which is equal to pi divided by root 2.
- Thus we have solved some problems on beta and gamma functions in this class.
- I hope you got an idea how to solve problems in this topic.
- Anyways, it is better to work on some more problems in this topic to get a better understanding.
- With this, we will end this session.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000066 · EV0000067 · EV0000068 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Problems on Beta and Gamma Function_continued transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/005_Problems_on_Beta_and_Gamma_Function_continued__5f7e0c86ddf2398a.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Beta and Gamma Function_continued transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/005_Problems_on_Beta_and_Gamma_Function_continued__5f7e0c86ddf2398a.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Beta and Gamma Function_continued transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/005_Problems_on_Beta_and_Gamma_Function_continued__5f7e0c86ddf2398a.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9982; only the JSON segmentation supplies the displayed timestamps.
Problems on Beta and Gamma Function_continued transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/005_Problems_on_Beta_and_Gamma_Function_continued__5f7e0c86ddf2398a.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
2. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
3. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.

4. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00018	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0044. Sidecar variants compare...
SRC00019	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0044. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00020	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0044. Sidecar variants compare...
SRC00021	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0044. Sidecar variants compare...

C Revision and retrieval practice